

Certification Specifications for Standard Changes and Repairs (AN)



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Training

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Let's go back in time.....

- Certification Frustrations

Great Products but how can I install them???

Individual Approvals being sought with high certification costs

Many Installing Agencies across Europe gaining the same Minor Changes for the same Equipment and Aircraft Types

Delays from EASA and NAA's getting STC's and Minor changes approved

To the Rescue!

- AML STC's

- CS-STAN

Issue 1 published in July 2015

Agenda

1. What is CS-STAN?
2. Who is responsible for Publishing CS-STAN?
3. CS-STAN Current Issue
4. Common Myths Surrounding CS-STAN
5. Subpart A – General
6. Subpart B Standard Changes
7. EASA / CAA / NAA Form 123
8. Common Myths Surrounding CS-STAN Answered
9. Using CS-STAN for Modification Flexibility
10. Future Issues – **We need Your Input!!!!**

1. What is CS-STAN?

Acceptable methods, techniques and practices for carrying out and identifying standard changes and standard repairs (SCs/SRs) as permitted pursuant to Part 21

CS STAN.00 Scope

SCs/SRs, designed in compliance with these certification specifications, are not subject to an approval process and, therefore, can be embodied in an aircraft when the conditions set out in the relevant points of Part 21 for SCs and SRs are met.

2. Who is Responsible for Publishing CS-STAN?



ED Decision 2022/009/R

The objective of this Decision is to reduce the regulatory burden and cost for general aviation (GA) with regard to the embodiment of specific Standard Changes (SCs) and Standard Repairs (SRs) in certain aircraft, while improving the level of safety.

The amendments introduced by this Decision are based on lessons learned and experience gained during the application of CS-STAN, proposals submitted by stakeholders, and on technological innovations introduced by the industry, which can bring safety benefits and allow the implementation of the latest technologies in a cost-efficient manner.

2. Who else is Responsible for Publishing CS-STAN?

NAA's outside of EASA

They will often use EASA's CS-STAN Verbatim but will have their own Independent Revision / Issue

3. CS-STAN Current Issue

The Current Issue for CS-STAN is Issue 4
Dated April 2022

**Note – Some NAA's may take time to adopt
a new Issue!!!!**

4. Common Myths Surrounding CS-STAN

- I don't need to do any paperwork to install equipment iaw CS-STAN
- The aircraft owner can fit the equipment using CS-STAN?

4. Common Myths Surrounding CS-STAN

- If I need to amend the Flight Manual or introduce an ICA, I cannot use CS-STAN, this needs to be an STC

4. Common Myths Surrounding CS-STAN

- CS-STAN is made available free of charge; therefore, I cannot charge the customer to use it

4. Common Myths Surrounding CS-STAN

- I cannot install equipment if it requires use of multiple CS-STAN Chapters

4. Common Myths Surrounding CS-STAN

- All CS-STAN Chapters have the same Applicability

4. Common Myths Surrounding CS-STAN

- All Equipment fitted under CS-STAN requires an EASA Form 1?

4. Common Myths Surrounding CS-STAN

- We are a Part 21 DOA so cannot / do not need to use CS-STAN

5. CS-STAN Subpart A - General

CS STAN.00	Scope.....
CS STAN.05	Embodiment of current SCs/SRs.....
CS STAN.10	Applicability.....
CS STAN.20	Operational Limitations or restrictions.....
CS STAN.30	Changes/Repairs that are not in conflict with TC holders' data
CS STAN.40	Referenced documents.....
CS STAN.42	Environmental conditions.....
CS STAN.47	Internal lithium batteries.....
CS STAN.48	Installation check flights
CS STAN.50	Instructions for continuing airworthiness (ICAs)
GM1 STAN.50	Instructions for continuing airworthiness (ICAs).....
CS STAN.60	Aircraft Flight Manual Supplement (AFMS)
GM1 STAN.60	Aircraft Flight Manual Supplement (AFMS).....
CS STAN.70	Acceptable Means of Compliance (AMC)
CS STAN.80	Definitions and abbreviations.....

CS-STAN.05 Embodiment of current SCs/SRs

As CS-STAN evolves, some SCs/SRs are amended, and their revision status is increased (e.g. CS-SC.XXXa is replaced with CS-SC.XXXb). In this respect, the new SC/SR version (in the example, CS-SC.XXXb) is the only one that is current and acceptable, and the old SC/SR version (in the example CS-SC.XXXa) becomes no longer acceptable for new embodiments.

Therefore, the following principles apply:

- It is mandatory to install the SC/SR using the latest revision that has been published.
- Always refer to the revision of the SC/SR in EASA Form '123'.

Note: A change or repair that was embodied against an older version of the relevant SC/SR, which was in force at the time of the embodiment, remains valid when a new revision of the SC/SR demands more conditions to be met, unless EASA mandates additional considerations by means of an airworthiness directive (AD).

PREAMBLE

ED Decision 2022/009/R

CS-STAN Issue 4

The following is a list of paragraphs affected by this Issue:

SUBPART A — GENERAL	
CS STAN.00	Amended (NPA 2021-06)
CS STAN.42	New (NPA 2021-06)
CS STAN.47	New (NPA 2021-06)
CS STAN.48	New (NPA 2021-06)
CS STAN.50	Amended (NPA 2021-06)
GM1 STAN.50	New (NPA 2021-06)
CS STAN.60	Amended (NPA 2021-06)
GM1 STAN.60	New (NPA 2021-06)
CS STAN.70	Amended (NPA 2021-06)
CS STAN.80	Amended (NPA 2021-06)
SUBPART B — STANDARD CHANGES	
LIST OF STANDARD CHANGES	Amended (NPA 2021-06)
CS-SC001	Amended (NPA 2021-06)
CS-SC002	Amended (NPA 2021-06)
CS-SC003	Amended (NPA 2021-06)
CS-SC004	Amended (NPA 2021-06)
CS-SC005	Amended (NPA 2021-06)
CS-SC006	New (NPA 2021-06)
CS-SC031	Amended (NPA 2021-06)
CS-SC032	Amended (NPA 2021-06)
CS-SC033	Amended (NPA 2021-06)
CS-SC034	Amended (NPA 2021-06)
CS-SC035	Amended (NPA 2021-06)
CS-SC036	Amended (NPA 2021-06)
CS-SC037	Amended (NPA 2021-06)
CS-SC038	Amended (NPA 2021-06)
CS-SC051	Amended (NPA 2021-06)
CS-SC052	Amended (NPA 2021-06)
CS-SC055	Amended (NPA 2021-06)
CS-SC056	Amended (NPA 2021-06)
CS-SC057	New (NPA 2021-06)
CS-SC059	New (NPA 2021-06)
CS-SC060	New (NPA 2021-06)
CS-SC061	New (NPA 2021-06)
CS-SC062	New (NPA 2021-06)
CS-SC081	Amended (NPA 2021-06)
CS-SC082	Amended (NPA 2021-06)
CS-SC083	Amended (NPA 2021-06)

CS-SC086	Amended (NPA 2021-06)
CS-SC087	New (NPA 2021-06)
CS-SC101	Amended (NPA 2021-06)
CS-SC102	Amended (NPA 2021-06)
CS-SC104	Amended (NPA 2021-06)
CS-SC105	Amended (NPA 2021-06)
CS-SC106	Amended (NPA 2021-06)
CS-SC107	Amended (NPA 2021-06)
CS-SC108	New (NPA 2021-06)
CS-SC109	New (NPA 2021-06)
CS-SC110	New (NPA 2021-06)
CS-SC152	Amended (NPA 2021-06)
CS-SC202	Amended (NPA 2021-06)
CS-SC203	Amended (NPA 2021-06)
CS-SC204	Amended (NPA 2021-06)
CS-SC206	Amended (NPA 2021-06)
CS-SC207	Amended (NPA 2021-06)
CS-SC208	New (NPA 2021-06)
CS-SC209	New (NPA 2021-06)
CS-SC210	New (NPA 2021-06)
CS-SC251	Amended (NPA 2021-06)
CS-SC252	New (NPA 2021-06)
CS-SC253	New (NPA 2021-06)
CS-SC401	Amended (NPA 2021-06)
CS-SC402	Amended (NPA 2021-06)
CS-SC403	Amended (NPA 2021-06)
SUBPART C — STANDARD REPAIRS	
LIST OF STANDARD REPAIRS	Amended (NPA 2021-06)
CS-SR801	Amended (NPA 2021-06)
CS-SR802	Amended (NPA 2021-06)
CS-SR804	Amended (NPA 2021-06)

ED Decision 2019/010/R

CS-STAN Issue 3

The following is a list of paragraphs affected by this Issue:

CS STAN.05	New (NPA 2018-10)
CS STAN.80	Amended (NPA 2018-10)
List of Standard Changes	Amended (NPA 2018-10)
CS-SC002	Amended (NPA 2018-10)
CS-SC003	Amended (NPA 2018-10)
CS-SC005	New (NPA 2018-10)
CS-SC034	Amended (NPA 2018-10)
CS-SC035	New (NPA 2018-10)
CS-SC036	New (NPA 2018-10)
CS-SC037	New (NPA 2018-10)

CS-SC038	New (NPA 2018-10)
CS-SC051	Amended (NPA 2018-10)
CS-SC052	Amended (NPA 2018-10)
CS-SC084	New (NPA 2018-10)
CS-SC085	New (NPA 2018-10)
CS-SC086	New (NPA 2018-10)
CS-SC105	New (NPA 2018-10)
CS-SC106	New (NPA 2018-10)
CS-SC107	New (NPA 2018-10)
CS-SC151	Amended (NPA 2018-10)
CS-SC152	Amended (NPA 2018-10)
CS-SC201	Amended (NPA 2018-10)
CS-SC206	New (NPA 2018-10)
CS-SC207	New (NPA 2018-10)
CS-SC401	Amended (NPA 2018-10)
List of standard repairs	Amended (NPA 2018-10)
CS-SR802	Amended (NPA 2018-10)

ED Decision 2017/014/R

CS-STAN Issue 2

The following is a list of paragraphs affected by this Issue:

CONTENTS	Amended (NPA 2016-17)
CS STAN.00	Amended (NPA 2016-17)
CS STAN.20	Amended (NPA 2016-17)
CS STAN.30	Amended (NPA 2016-17)
CS STAN.80	Amended (NPA 2016-17)
List of Standard Changes	Amended (NPA 2016-17)
CS-SC002	Amended (NPA 2016-17)
CS-SC003	Amended (NPA 2016-17)
CS-SC031	Amended (NPA 2016-17)
CS-SC032	New (NPA 2016-17)
CS-SC033	New (NPA 2016-17)
CS-SC034	New (NPA 2016-17)
CS-SC051	Amended (NPA 2016-17)
CS-SC052	Amended (NPA 2016-17)
CS-SC053	Amended (NPA 2016-17)
CS-SC054	Amended (NPA 2016-17)
CS-SC055	Amended (NPA 2016-17)
CS-SC056	Amended (NPA 2016-17)
CS-SC058	New (NPA 2016-17)
CS-SC081	New (NPA 2016-17)
CS-SC083	New (NPA 2016-17)
CS-SC101	Amended (NPA 2016-17)
CS-SC102	New (NPA 2016-17)
CS-SC103	New (NPA 2016-17)

CS-SC104	New (NPA 2016-17)
CS-SC153	Amended (NPA 2016-17)
CS-SC202	Amended (NPA 2016-17)
CS-SC203	Amended (NPA 2016-17)
CS-SC205	New (NPA 2016-17)
CS-SC251	Amended (NPA 2016-17)
CS-SC401	Amended (NPA 2016-17)
CS-SC402	Amended (NPA 2016-17)
CS-SC403	New (NPA 2016-17)
List of standard repairs	Amended (NPA 2016-17)
CS-SR802	Amended (NPA 2016-17)
CS-SR803	New (NPA 2016-17)
CS-SR804	New (NPA 2016-17)

CS-STAN.10 Applicability

In addition to the conditions of 21.A.90B and 21.A.431B, for each SC/SR, these Certification Specifications may further restrict its applicability to certain aircraft, or to some areas of an aircraft, or to certain aircraft operations.

21.A.90B Standard changes

Standard changes are changes to a type-certificate:

1. in relation to:

- (i) aeroplanes of 5 700 kg Maximum Take-Off Mass (MTOM) or less;
- (ii) rotorcraft of 3 175 kg MTOM or less;
- (iii) sailplanes, powered sailplanes, balloons and airships, as defined in ELA1 or ELA2

CS-STAN.10 Applicability

CS-Scoo1b – Installation of VHF Voice Communication Equipment

Aeroplanes not being complex motor-powered aircraft with a maximum cruising speed in ISA conditions below 250 kt IAS, rotorcraft that are not complex motor-powered aircraft, and any ELA2 aircraft.

CS-SC101c – Installation of ELT / PLB

Aeroplanes with a MTOM below 2 730 kg, rotorcraft that are not complex motor-powered aircraft, with a MTOM below 1 200 kg and with 4 occupants or fewer, and any ELA2 aircraft.

CS-SCoo3d – Installation of Audio Selector Panels and Amplifiers

This SC is applicable to aeroplanes and to rotorcraft that are not complex motor-powered aircraft, and to any ELA2 aircraft.

CS-STAN.20 Operational Limitations or restrictions

SCs/SRs, as described in these certification specifications, may contain operational limitations or restrictions with regard to the use of an aircraft instrument/equipment.

Equipment installed as part of an SC cannot be used to eliminate or reduce the existing airworthiness limitations and operational limitations of the aircraft (e.g. an aircraft certified only for VFR operation cannot be authorised to operate IFR as a result of modifications embodied through CS-STAN).

CS-STAN.42 Environmental Conditions

If the equipment is required to meet the environmental conditions to be expected during normal operations, the following provides appropriate guidance:

— Ensure that the equipment manufacturer has declared that the equipment is suitable for installation on a specific aircraft type and for the operations defined in the standard change (SC). This statement could be included in a document (e.g. a DDP, CMM, etc.) issued by the equipment manufacturer, or in a written communication (e.g. email).

CS-STAN.48 Installation Check flights

An installation check flight is considered part of the SC installation activity, and it can be conducted without an individual permit to fly or any other certificate of release to service for the SC installation, as long as there is no other ongoing maintenance activity on the aircraft (i.e. other maintenance, which has not yet been released to service).

CS-STAN.50 Instructions for continuing airworthiness (ICA's)

Depending on the SC/SR being embodied, the aircraft instructions for continuing airworthiness (ICAs) may need to be updated.

This update is considered part of this SC/SR and, therefore, requires no specific approval. A template that could be used for the supplement to the ICAs is provided in GM CS STAN.50.

	ICAs	Doc No:
		Revision:
		Date:
Company approval reference, if any	Standard Change / Standard Repair reference	Page:

AIRCRAFT TYPE

SUPPLEMENT TO THE INSTRUCTIONS FOR CONTINUING AIRWORTHINESS (ICAs)

LOG OF REVISIONS

<i>Edition</i>	<i>Revision</i>	<i>Date</i>	<i>Description</i>	<i>Page</i>

0 — Introduction

These instructions provide information to maintain and inspect the product to which they refer, installed on the specified aircraft.

These instructions are provided with each system/equipment, and will be provided to any owner of the system/equipment on request.

1 — Instructions for continuing airworthiness (ICAs)

The ICAs include:

- 1) a list of parts installed;
- 2) scheduled maintenance (maintenance programme);
- 3) troubleshooting;
- 4) the shelf life of the component(s);

1.1 List of parts installed

Item	Part number	Description	Quantity

1.2 Maintenance programme

Specify a maintenance programme that includes the frequency and extent of the maintenance tasks necessary to ensure the continued airworthiness of the aircraft.

Task reference number (possibly referring to ATA chapter)	Description (high level)	Interval (flight hours/cycles or calendar interval)	Reference to detailed maintenance instructions in Chapter 2	Remarks

1.3 Troubleshooting

If available, troubleshooting information describing probable malfunctions, how to recognise them, and the remedial action(s) for those malfunctions should be provided.

1.4 Shelf life of the component(s)

If the shelf life is provided by the component manufacturer, add the relevant information.

2 — Maintenance tasks

Insert here all maintenance tasks; each of them should list the following: the tools, the purpose of inspections/maintenance, and the operations to be performed.

3 — Vendor instructions (OEM)

If available, insert here a reference to the maintenance documentation released by the equipment manufacturer or the vendor.

Item	CMM

4 — Wiring diagram(s)

If applicable, insert here new or modified wiring diagram(s).

CS-STAN.6o Aircraft Flight Manual Supplement

Depending on the SC/SR being embodied, the AFM may need to be updated.

The supplement to the AFM is considered part of this SC/SR and, therefore, requires no specific approval. A template that could be used for the supplement to the AFM is provided in GM1 STAN.6o.

	AFMS	Doc No:
		Revision:
		Date:
Company approval reference, if any	Standard Change / Standard Repair reference	Page:

TITLE

A/C :

Version :

S/N :

Registration :

This Aircraft Flight Manual Supplement (AFMS) must be attached to the approved AIRCRAFT FLIGHT MANUAL (AFM) kept on board the aircraft.

The information contained in this document integrates and, in some cases, supersedes the information reported in the corresponding section of the approved AFM.

LIST OF EFFECTIVE PAGES

Page	Edition — Revision — Date	Page	Edition — Revision — Date

LIST OF EDITION/REVISION INCORPORATED

Edition	Revision	Date	Description	Page

6. CS-STAN Subpart B – Standard Changes

The list of Standard Changes are grouped as follows:

Group Systems - Communication

- Electrical

- Avionics/Nav/Instruments

- Mechanical and structural

Group Cabin

Group Survivability Equipment

Group Powerplant and fuel systems

Group Flight

Group Miscellaneous

SUBPART B — STANDARD CHANGES

LIST OF STANDARD CHANGES

Group Systems — Communication:

- CS-SC001b — Installation of VHF voice communication equipment
- CS-SC002d — Installation of Mode S elementary surveillance equipment
- CS-SC003d — Installation of audio selector panels and amplifiers
- CS-SC004b — Installation of antennas
- CS-SC005b — Installation of an ADS-B OUT system combined with a transponder system
- CS-SC006a — Exchange of a COM, NAV or NAV/COM unit for a combined VHF voice communication and navigation (NAV/COM) unit

Group Systems — Electrical:

- CS-SC031c — Exchange of conventional anti-collision lights, position lights, and landing and taxi lights for LED-type lights
- CS-SC032c — Installation of anti-collision lights
- CS-SC033b — Installation of cabin and cockpit conventional lights for LED-type lights
- CS-SC034c — Exchange of an existing battery for a lithium iron phosphate (LiFePO₄) battery system
- CS-SC035b — Installation of solar cells on sailplanes
- CS-SC036b — Installation of visual awareness lights
- CS-SC037b — Exchange of a main aircraft battery
- CS-SC038b — Installation of DC-to-DC converters

Group Systems — Avionics/NAV/Instruments:

- CS-SC051d — Installation of 'FLARM' equipment
- CS-SC052d — Installation of GNSS equipment
- CS-SC053b — Installation of radio marker receiving equipment
- CS-SC054b — Exchange of distance-measuring equipment (DME)
- CS-SC055c — Exchange of ADF equipment
- CS-SC056c — Installation of VOR equipment
- CS-SC057a — Installation of an electronic conspicuity (EC) function
- CS-SC058a — Installation of traffic awareness beacon system (TABS) equipment
- CS-SC059a — Installation of a gyroscopically stabilised direction indicator

CS-SC060a — Installation of a secondary attitude indicator

CS-SC061a — Installation of an aircraft tracking system

CS-SC062a — Installation of an awareness function or awareness device

Group Systems — Mechanical and structural:

CS-SC081b — Exchange of tyres (inner tubes/outer tyres)

CS-SC082b — Exchange of skids on wing tips/fuselage tails

CS-SC083b — Exchange of flexible seals on control surfaces

CS-SC084a — Repainting of composite aircraft structures

CS-SC085a — Exchange of an aircraft livery paint and decorative sticker scheme

CS-SC086b — Exchange of a balloon bottom end

CS-SC087a — Installation of balloon spare parts

Group Cabin:

CS-SC101c — Installation of emergency locator transmitter (ELT) equipment / satellite personal locator beacon

CS-SC102b — Installation of DC power supply systems (DC-PSSs) for portable electronic devices (PEDs)

CS-SC103a — Exchange of interior material covering floor, sidewall and ceiling

CS-SC104b — Installation of lightweight in-flight recording systems

CS-SC105b — Installation of mounting systems to hold equipment

CS-SC106b — Installation of flight-time recorders

CS-SC107b — Installation of carbon monoxide (CO) detectors

CS-SC108a — Exchange of handheld fire extinguishers for halon-free types

CS-SC109a — Installation of handheld fire extinguishers

CS-SC110a — Installation of annunciators

Group Survivability Equipment:

CS-SC151b — Installation of headrests

CS-SC152c — Changes to seat cushions, including the use of alternative foam materials

CS-SC153b — Exchange of safety belts — torso restraint systems

Group Powerplant and fuel systems:

CS-SC201b — Exchange of powerplant instruments

CS-SC202c — Use of aviation gasoline (Avgas) UL 91

CS-SC203c — Use of aviation gasoline (Avgas) Hjelmcø 91/96 UL and 91/98 UL

CS-SC204b — Installation of an externally powered engine preheater

CS-SC205a — Installation of fuel low-level sensor (FLLS)

CS-SC206b — Exchange of fixed-pitch wooden propellers

CS-SC207b — Exchange of the fuel cylinders on hot-air balloons

CS-SC208a — Installation of a multifunction display for powerplant instruments

CS-SC209a — Exchange of a propeller governor

CS-SC210a — Installation of a fuel flow/pressure instrument

Group Flight:

CS-SC251c — Installation of an angle-of-attack (AoA) indicator system

CS-SC252a — Installation of a tactile stall-warning indicator system

CS-SC253a — Installation of a device receiving uplinked weather radar information

Group Miscellaneous:

CS-SC401d — Installation of basic flight instruments

CS-SC402c — Installation of sailplane equipment

CS-SC403b — Provisions for the installation of lightweight cameras

[Issue: STAN/4]

6. CS-STAN Subpart B – Standard Changes

Each Standard Change Chapter has the following Sections:

1. Purpose
2. Applicability / Eligibility
3. Acceptable Methods, techniques, and practices
4. Limitations
5. Manuals
6. Release to Service

Standard Change CS-SC003d

INSTALLATION OF AUDIO SELECTOR PANELS AND AMPLIFIERS

1. Purpose

This SC is for the installation of audio selector panels and amplifiers.

Audio selector amplifiers that feature automatic speech recognition are eligible for installation by means of this SC only if the automatic speech recognition feature is disabled.

2. Applicability/Eligibility

This SC is applicable to aeroplanes and to rotorcraft that are not complex motor-powered aircraft, and to any ELA2 aircraft.

3. Acceptable methods, techniques, and practices

The following standards contain acceptable data:

- FAA Advisory Circular AC 43-13-2B, Chapter 2; and
- FAA Advisory Circular AC 43.13-1B, Chapter 11 or ASTM F2639-18 or subsequent revisions.

Additionally, the following conditions apply:

- the equipment is authorised in accordance with ETSO-C50c or ETSO-C139a, or later amendments, or equivalent standards;
- the equipment has at least the audio functionality of the previously installed equipment, and is compatible with the existing installation;
- the equipment is compatible with the connections to the existing communication and navigation systems;
- the equipment is suitable for the environmental conditions to be expected during normal operations; see CS STAN.42 in Subpart A for guidance; and
- the instructions and tests defined by the equipment manufacturer are followed.

4. Limitations

Any limitations defined by the equipment manufacturer apply.

Any limitations of the existing installation remain valid.

In the case of aircraft approved for NVIS/NVG, the change cannot be considered a SC.

5. Manuals

Amend the AFM with an AFMS that contains or references the equipment instructions for operation, as required.

If the audio selector includes connections to equipment with functions that are not part of the aircraft's control system or navigation system, and which may emit distracting sounds, add a limitation in the AFMS stating that these functions shall not be used during take-off and landing. Examples of such functions include telephony, personal messaging functions and music players.

Amend the ICAs to establish maintenance actions/inspections and intervals, as required.

6. Release to service

This SC is not suitable for the release to service of the aircraft by the pilot-owner.

[Issue: STAN/2]

[Issue: STAN/3]

[Issue: STAN/4]

7. EASA / CAA / NAA Form 123

The EASA / CAA / NAA Form 123 is the record of the Embodiment of the Standard Change / Standard Repair that is passed to the owner on completion of the Standard Change.

This form is not the Release to Service

The owner is required to sign Section 12 and the installer should keep a copy for their records.

EASA Form 123 — Standard Change/Standard Repair (SC/SR) embodiment record

EASA Form 123 — Standard Change/Standard Repair (SC/SR) embodiment record		1 SC/SR number(s):	
2 SC/SR title & description:			
3 Applicability:			
4 List of parts (description/Part-No/Qty):			
5 Operational limitations/affected aircraft manuals. Copies of these manuals are provided to the aircraft owner:			
6 Documents used for the development and embodiment of this SC/SR:			
* Copies of the documents marked with an asterisk are handed to the aircraft owner.			
7 Instructions for continuing airworthiness. Copies of these manuals are provided to the aircraft owner:			
8 Other information:			
9a ☐ This SC complies with the criteria established in 21A.90B(a) and with the relevant paragraphs of CS-STAN.			
9b ☐ This SR complies with the criteria established in 21A.431B(a) and with the relevant paragraphs of CS-STAN.			
10 Date of SC/SR embodiment:		11 Identification data and signature of the person responsible for the embodiment of the SC/SR:	
12 Signature of the aircraft owner. This signature attests that all relevant documentation is handed over from the issuer of this form to the aircraft owner, and, therefore, the latter becomes aware of any impact or limitations on operations or additional continuing airworthiness requirements which may apply to the aircraft due to the embodiment of the change/repair.			

CAA Form 123: Standard Change/Standard Repair (SC/SR) embodiment record



1. SC/SR number(s):	
2. SC/SR title & description:	
3. Applicability:	
4. List of parts (Part-No./Description/Qty):	
5. Operational limitations/affected aircraft manuals. Copies of these manuals are provided to the aircraft owner:	
6. Documents used for the development and embodiment of this SC/SR: *Copies of the documents marked with an asterisk are handed to the aircraft owner.	
7. Instructions for continuing airworthiness. Copies of these manuals are provided to the aircraft owner:	
8. Other information:	
9. a. This SC complies with the criteria established in 21A.90B(a)¹ or CAP1419 Para. 7.2¹ and with the following chapter(s) of Certification Specifications CS-STAN. b. This SR complies with the criteria established in 21A.431B(a)¹ or CAP1419 Para. 7.2¹ and with the following chapter(s) of Certification Specifications CS-STAN.	
10. Date of SC/SR embodiment:	11. Identification data and signature of the person responsible for the embodiment of the SC/SR:
12. Signature of the aircraft owner. This signature attests that all relevant documentation is handed over from the issuer of this form to the aircraft owner, and, therefore, the latter becomes aware of any impact or limitations on operations or additional continuing airworthiness requirements which may apply to the aircraft due to the embodiment of the modification/repair. Date:	

¹ Delete as applicable

8. Common Myths Surrounding CS-STAN Answered

- I don't need to do any paperwork to install equipment iaw CS-STAN

Each Chapter within CS-STAN lists the Applicability, Acceptable Methods & Limitations. Embodiment of a Chapter must be recorded using EASA/CAA Form 123 and an entry made in the Aircraft Log Book.

8. Common Myths Surrounding CS-STAN Answered

- The aircraft owner can fit the equipment using CS-STAN?

6. Release to service

This SC is not suitable for the release to service of the aircraft by the pilot-owner.

8. Common Myths Surrounding CS-STAN Answered

- If I need to amend the Flight Manual or introduce an ICA, I cannot use CS-STAN, this needs to be an STC

5. Manuals

Amend the AFM with an AFMS that contains or references the equipment instructions for operation, as required.

Amend the ICAs to establish maintenance actions/inspections and intervals, as required.

8. Common Myths Surrounding CS-STAN Answered

- CS-STAN is made available free of charge; therefore, I cannot charge the customer to use it

CS-STAN must be documented iaw the applicable Chapter. Drawings and a Modification datapack (Including an ICA and AFMS / RFMS) need to be produced and supplied to the customer along with an EASA/CAA Form 123

8. Common Myths Surrounding CS-STAN Answered

- I cannot install equipment if it requires use of multiple CS-STAN Chapters

Standard Change CS-SCo52d (Installation of GNSS Equipment)

For installations that also provide voice communications functionality and/or navigation capability, CS-SCo52() may be applied concurrently with CS-SCo01(), and/or CS-SCo06() and/or CS-SCo56().

8. Common Myths Surrounding CS-STAN Answered

- All CS-STAN Chapters have the same Applicability

Each Chapter Independently lists the Applicability/Eligibility

CS-Scoo1b – Installation of VHF Voice Communication Equipment

Aeroplanes not being complex motor-powered aircraft with a maximum cruising speed in ISA conditions below 250 kt IAS, rotorcraft that are not complex motor-powered aircraft, and any ELA2 aircraft.

CS-SC101c – Installation of ELT / PLB

Aeroplanes with a MTOM below 2 730 kg, rotorcraft that are not complex motor-powered aircraft, with a MTOM below 1 200 kg and with 4 occupants or fewer, and any ELA2 aircraft.

8. Common Myths Surrounding CS-STAN Answered

- All Equipment fitted under CS-STAN requires an EASA Form 1?

ED Decision 2022/009/R

EASA also specifies in CS-STAN Issue 4, when applicable and justified, the acceptance of parts without an EASA Form 1 in accordance with point 21.A.307(b)(4) of Commission Implementing Regulation (EU) 2021/699, which shall apply from 18 May 2022.

8. Common Myths Surrounding CS-STAN Answered

- We are a Part 21 DOA so cannot / do not need to use CS-STAN

CS-STAN does not identify who can use it

CS-STAN Documentation can sit outside of a Part 21 DOA's normal procedures

9. Using CS-STAN for Modification Flexibility

- Example: Installing a Traffic System onto a Part-27 Rotorcraft. No AML STC exists, therefore it requires either Part 21 DOA or Minor Change Approval via EASA / NAA.

You want to make the Modification a Series Approval to cover all types on the TCDS.

Or even extend it and have an MML.

What is typically the limiting factor?

Antenna Location!!!

9. Using CS-STAN for Modification Flexibility

Use CS-STAN Chapter CS-SC004b “Installation of Antennas” in conjunction with the Series Modification.

Make a statement in the Series Modification Datapack that “The installation of associated Antennas are covered under CS-STAN Chapter CS-SC004b (or subsequent issue/revision) “Installation of Antennas”

Using CS-STAN for Antenna Approval in a Minor Change Application will increase the chances of a MML being accepted by the Authority



Questions?

10. What will be added in CS-STAN Issue 5 or Future Issues?

We need Your Input!!!!

CS-SC101c – “Installation of a ELT / PLB” –
Request to expand this to non complex motor-
powered aircraft for both Fixed Wing and
Rotorcraft

New Chapter for installation of Rad Alt
System

CS-SC054b “Exchange of distance-measuring
equipment (DME)” to be changed to
installation instead of exchange

CS-SC055c “Exchange of ADF equipment” to
be changed to installation instead of
exchange